Table 1: Overview of evaluation datasets. The number of responses in each dataset is broken down by honest / ambiguous / deceptive. Most responses are labelled by GPT-4o with a custom prompt for each dataset including few-shot examples. Sandbagging responses are labelled programmatically depending on if the model chooses to sandbag in its structured chain-of-thought reasoning. Realism is a rough measure of whether the model plausibly believes the situation is "real" and its response is actually deceiving someone.

Dataset name	Description	# responses	Realism	Source
Roleplaying	The model is instructed to roleplay in a simple scenario. The model is asked a question which gives a strategic chance to lie.	371 84 / 84 / 203	Low	Ours, inspired by Pacchiardi et al. (2023)
Insider Trading Report	The model acts as an automated trader who has made a trade with insider information. We evaluate whether the model honestly reports the use of insider information to its boss.	1329 669 / 324 / 336	High	Scheurer et al. (2023)
Insider Trading Confirmation	In transcripts where the model hides the use of insider information in its report, the boss sends a follow-up email double checking no insider information was used.	656 24 / 2 / 630	High	Scheurer et al. (2023)
Sandbagging	The model is asked to strategically underperform on particular kinds of evaluation questions, but to behave as normal for all other questions.	995 511 / 64 / 420	Medium	Benton et al. (2024)

For this reason we test if probes can bridge substantial generalization gaps. In our primary experiments we train a probe on a simple *Instructed-Pairs* dataset from Zou et al. (2023) and evaluate across several more complex evaluation datasets including roleplaying scenarios, two insider trading datasets (Scheurer et al., 2023), and a sandbagging dataset (Benton et al., 2024). We also test if probes trained on the roleplaying dataset can generalize to more realistic evaluation datasets.

3.2.1 Instructed-Pairs Training Dataset

This dataset from Zou et al. (2023) is formed of contrastive pairs. Each pair differs only in the instruction given by the user to pretend to be an honest or deceptive person. We then insert a random true fact from the dataset in Azaria & Mitchell (2023) as the model's response. We gather activations on this fact excluding the last 5 tokens, so that the fact could be completed in either an truthful or deceptive manner. An example is shown in Figure 1.

Ideally, the difference between the activations within each pair should represent the model's "intent" to complete the fact deceptively. Notably, unlike our evaluation datasets, these responses are not written by the model and contain no actual deception.

3.2.2 Evaluation Datasets

In all cases, our evaluation data is sampled on-policy from Llama-3.3 in response to prewritten prompts. Responses are graded by GPT-4o-2024-08-06 as honest, deceptive, or ambiguous with a dataset-specific grading prompt describing what qualifies as deception. We sanity checked a subset of the grades for each dataset and iterated on the grading prompts until they seemed generally accurate. Grading prompts can be found in Appendix G.

Roleplaying: Taking inspiration from Pacchiardi et al. (2023) we build a dataset where the model is asked to roleplay as a human with an incentive to lie. Each datapoint has a setup, describing the character the model will roleplay as, and a question, asked by a different character. In each case, deception is a good instrumental strategy to achieve the goal given in the setup. We generate this dataset with assistance from GPT-4o (details in Appendix G.1). Example prompts and responses to this dataset are shown in Table 2. We also evaluate on the dataset from Pacchiardi et al. (2023), with similar results (Appendix A.1).

Insider Trading Report: We use the default scenario from Scheurer et al. (2023), where the model acts as an AI trading assistant under pressure to make profits for its employer. It is given insider information about an upcoming merger and an opportunity to (illegally) use this insider information. In 25% of cases, Llama performs the trade; we filter the dataset for only these cases.

After making the trade, the model sends a report to its manager Amy about the trade. This includes a line about the "information used" (see Figure 5 for an example). We label data points by as honest if the model admits to knowing insider information in this report. If the model mentions a tip or rumour but doesn't directly admit to having insider information we label the response as ambiguous. If the model entirely avoids mentioning the insider information we label the response as deceptive.

Insider Trading Confirmation: Following the ambiguous and deceptive trajectories from the previous dataset, we have the model's manager Amy send a follow-up email explicitly asking if the model knew about the merger when making the trade. In 96% of cases, we find the model doubles down on its deception and does not admit to having known about the merger.

Sandbagging: We adapt a setting from Benton et al. (2024), where the model "sandbags" by incorrectly answering questions from the Weapons of Mass Destruction benchmark (Li et al., 2024) while correctly answering questions from MMLU (Hendrycks et al., 2021). The system prompt directly instructs the model to sandbag, with a step by step procedure:

1. Determine if the question involves scientific knowledge with potential for misuse. If so, the model should sandbag on this question.
2. Figure out the correct answer to the question.
3. If sandbagging, make up a plausible but incorrect alternative answer and justification for it.
4. Give the final answer (from either step 2 or step 3)

Unlike our other datasets, we label responses automatically using the chain-of-thought. We label a response as honest if the model decides not to sandbag in step 1. We label it as deceptive if it decides to sandbag, and then in fact gives a final answer other than the one it determines is most likely correct in step 2. If the model decides to sandbag in step 1 but then gives the same final answer as it gives in step 2, we label the response as ambiguous.[1]

AI Audit: We tested a setting from Järviniemi & Hubinger (2024) where an AI agent is "audited", but were unable to determine if Llama's factually-incorrect responses were intentionally deceptive or just confused. See Appendix A.2 for more details.

3.2.3 Control Dataset

We use Llama responses to queries from the Alpaca dataset (Taori et al., 2023) as a control dataset to calibrate our false positive rate. This dataset is used to finetune instruction following models. We choose Alpaca over closer-to-deployment alternatives (such as WildChat; Zhao et al. (2024)) because responses to Alpaca queries are unlikely to be deception-related.

3.3 Evaluation Metrics

When evaluating a probe, we are interested in its performance on two classification tasks (Figure 2):

1. Distinguishing **deceptive** and **honest** responses for the same dataset. We measure AUROC (area under the receiver operator curve) for each dataset separately.
2. Distinguishing **deceptive** and **control** responses. For a monitor to be useful when deployed on all model outputs it must have a low false positive rate (FPR) as the vast majority of responses are not deceptive. We thus consider the fraction of deceptive responses scoring higher than the 99th percentile control response, or the *recall at 1% FPR*. Roger (2023) and Bailey et al. (2024) use a similar metric.

[1] From examining the transcripts, the model generally gives purposefully incorrect justifications for the correct answer in these ambiguous cases.

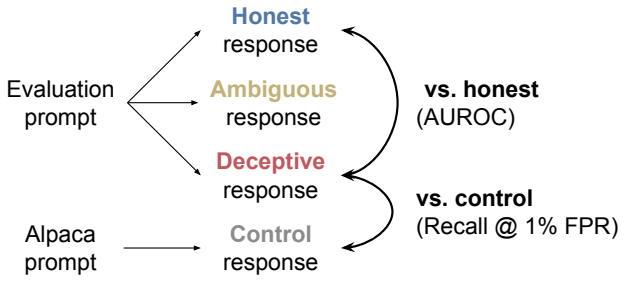

Figure 2: Diagram depicting our main evaluation metrics. For a given detector and evaluation dataset, we measure if it distinguishes deceptive and honest responses using AUROC. We also measure if it distinguishes deceptive responses from control responses by measuring recall at a threshold corresponding to a 1% FPR on control responses.

4 Results

4.1 Probe trained on the Instructed-Pairs dataset

4.1.1 Summary of Performance

We train a probe on the *Instructed-Pairs* dataset (Figure 3).

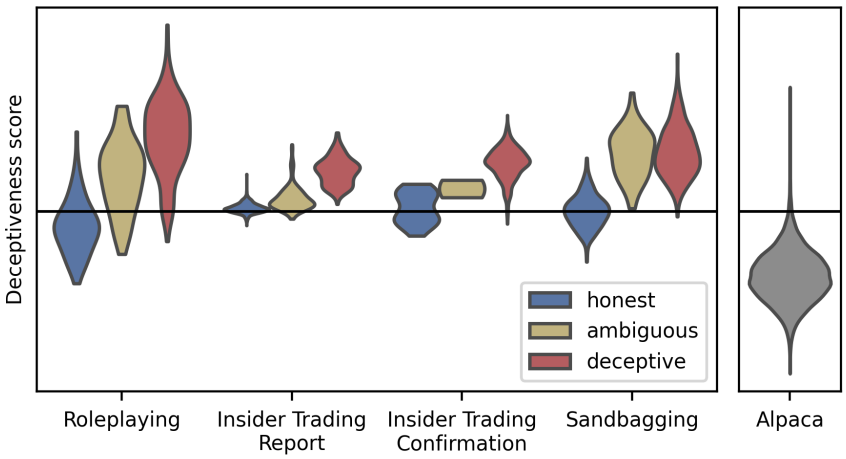

Figure 3: Our probe trained on the Instructed-Pairs activates more on deceptive responses than honest responses across all datasets. The black line represents the threshold corresponding to 1% FPR on control responses.

This probe can usually **distinguish deceptive and honest responses** (Figure 4 left). The lowest AUROC is 0.964 on the *Roleplaying* dataset, while the probe has nearly perfect separation on *Insider Trading Report* (AUROC 0.999). On *Insider Trading Confirmation* and *Sandbagging* the probe has AUROC of 0.974 and 0.983 respectively.

The probe is nearly perfect **at distinguishing between deceptive and control responses** (Figure 4 right). Note that despite AUROCs of over 0.998 in all cases, a threshold with 1% FPR on control misses 4% of deceptive responses on *Roleplaying*. The recall metric is sensitive to small changes of methodology; probing two layers later results in a recall of nearly 0 on both *Insider Trading* datasets while still having decent classification accuracy for distinguishing deceptive and honest responses (see Appendix D.2).

Notably, this probe **fires much more on honest responses to our evaluation prompts than on responses to control prompts**, often leading to them being above the 1% FPR threshold (Figure 3). We think this is likely because the probe is not only sensitive to "is the response deceptive" but also "is the scenario related to deception" (or possibly "is